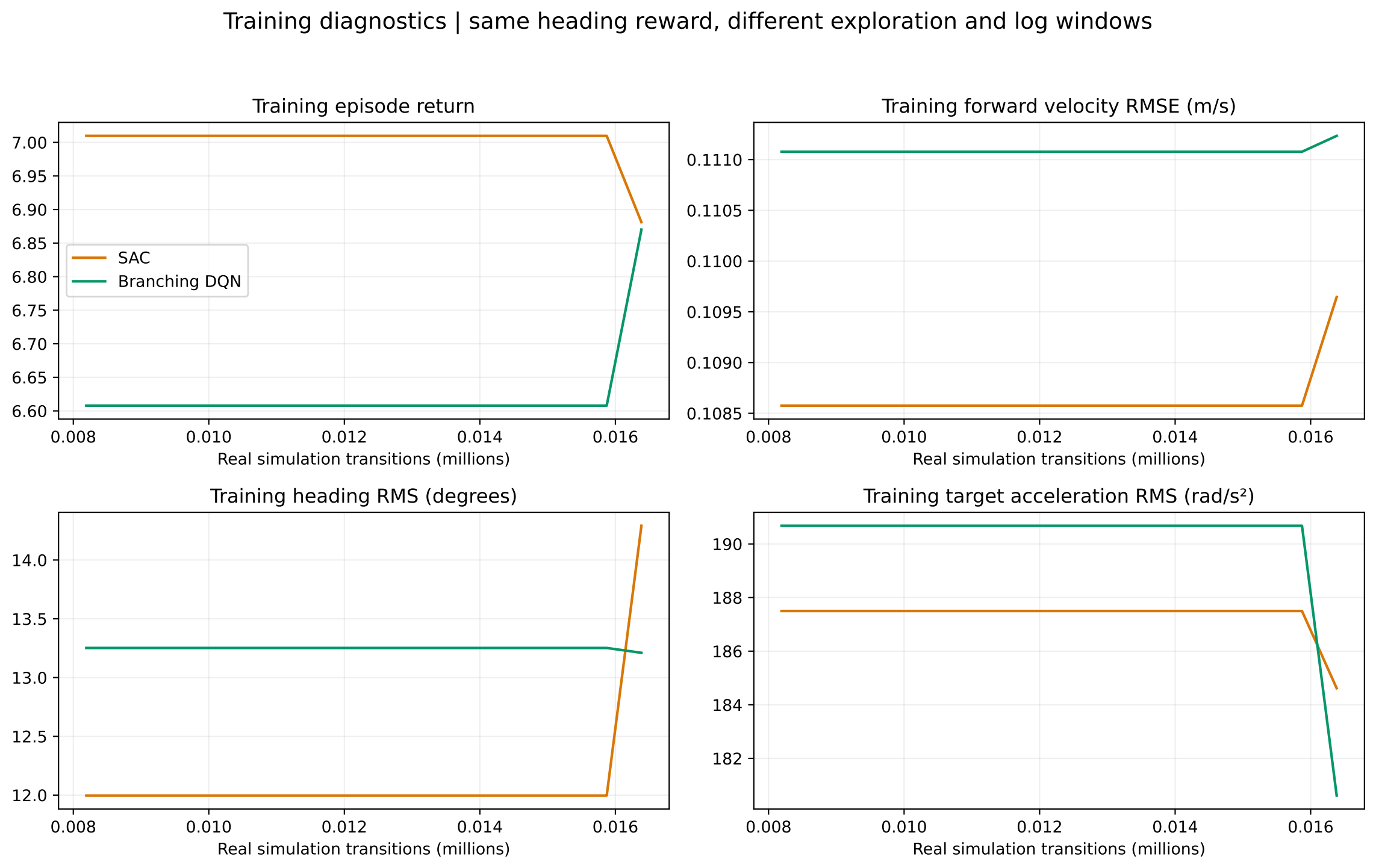
Training diagnostics | same heading reward, different exploration and log windows
Training episode return
7.00
6.95
6.90
6.85
6.80
6.75
6.70
6.65
6.60
SAC
Branching DQN
0.008
0.010
0.012
0.014
0.016
Real simulation transitions (millions)
Training forward velocity RMSE (m/s)
0.1110
0.1105
0.1100
0.1095
0.1090
0.1085
0.008
0.010
0.012
0.014
0.016
Real simulation transitions (millions)
Training heading RMS (degrees)
14.0
13.5
13.0
12.5
12.0
0.008
0.010
0.012
0.014
0.016
Real simulation transitions (millions)
Training target acceleration RMS (rad/s²)
190
188
186
184
182
0.008
0.010
0.012
0.014
0.016
Real simulation transitions (millions)
Faint: individual seeds. Bold/band: mean/sample SD at shared steps. PPO return window 100 episodes; SAC/DQN 256.